Theory of Elementary Particles: Lecture 8

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Quark color
- Δ^{++}

Hadron wave function

Nucleon Wave Function

Magnetic moment of nucleon

Deep Inelastic Scattering and Parton Model

Electron-proton scattering
- Hadronic Tensor
- Structure Functions
- Bjorken scaling

Parton Model
- Callan-Gross relation

Parton Distribution Functions

Quark color

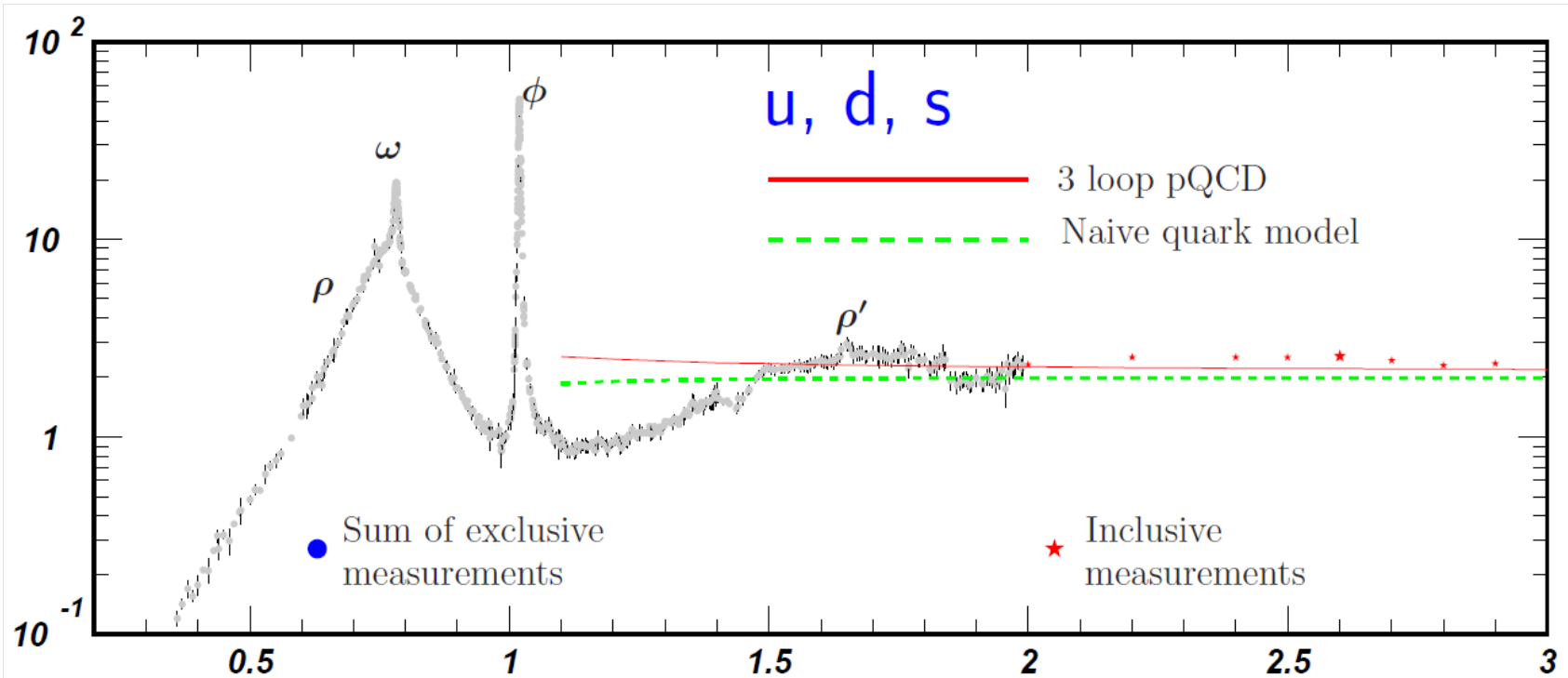
Arguments supporting $SU(3)$ for the color symmetry.

Δ^{++}

- $3/2$-spin particle
$$\Delta^{++} =\mid u\, u\, u\rangle$$
belongs to the totally symmetric decuplet $\mathbf{10}$!
- Consider
$$\Delta^{++}(J_3 = \frac{3}{2}) =\mid u\uparrow u\uparrow u\uparrow\rangle$$
- if one assumes that space part of the wave function is S-wave (good predictions of the magnetic moments of baryons), then the function is symmetric, but Baryon is fermion and it should be totally anti-symmetric wave function!
- Two quarks (fermions) can not occupy the same state according to the Pauli exclusion principle
- **One assumes existence of the hidden degrees of freedom for quarks colour in order to distinguish three quarks**
$$\Delta^{++}(J_3 = \frac{3}{2}) = \frac{1}{\sqrt{6}}\epsilon_{ijk} \mid u^i\uparrow u^j\uparrow u^k\uparrow\rangle \qquad (1)$$
$i, j, k = 1, 2, 3$ colors
- $SU(3)$ color symmetry

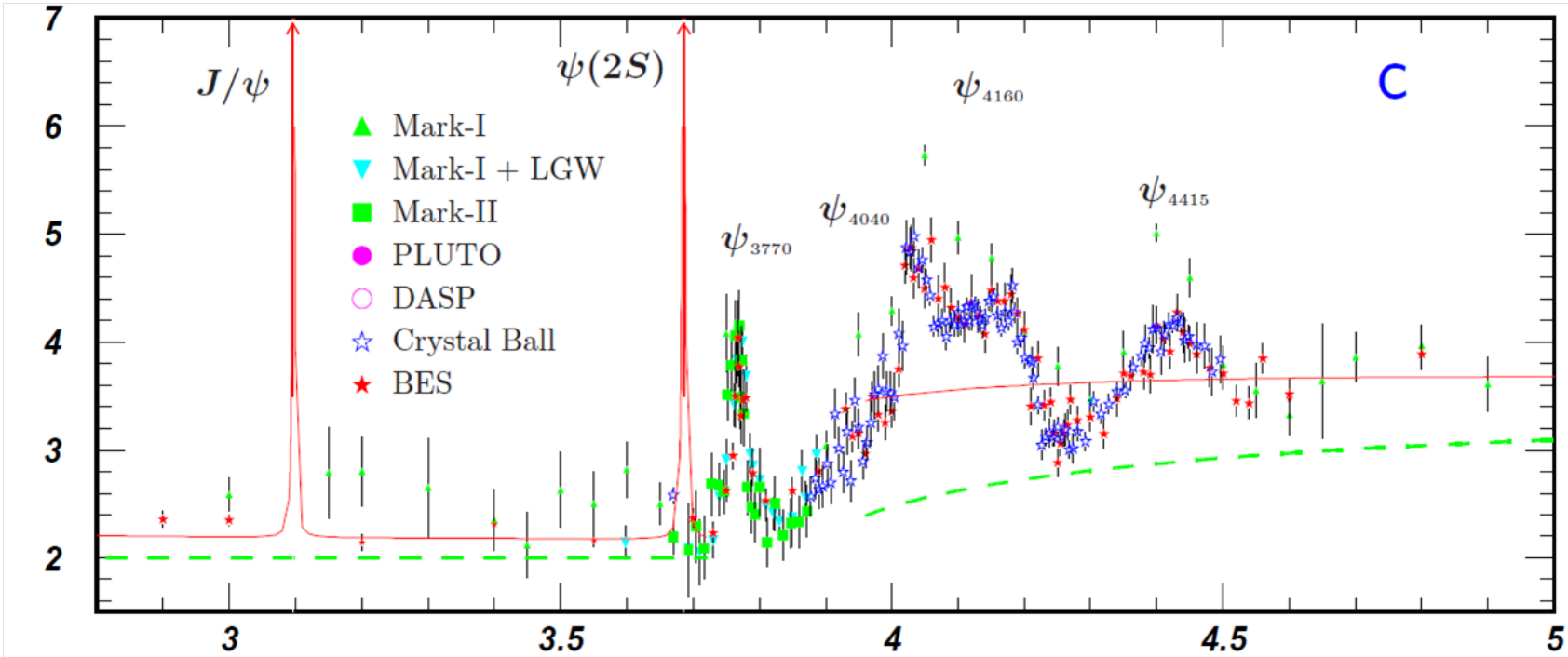
e^+e^- scattering

- $$e^+e^- \to q\bar{q} \to hadrons$$
- $$\sigma(e^+e^- \to hadrons) = \frac{4\pi\alpha^2}{3s} N_c \sum_{i=1}^{N_f} Q_i^2$$
- Let us define
$$R = \frac{\sigma(e^+e^- \to hadrons)}{\sigma(e^+e^- \to \mu^+\mu^-)} = N_c \sum_{i=1}^{N_f} Q_i^2$$
- below $\sqrt{s} = 3$ GeV only u, d and s quarks contribute
$$R = (N_c = 3)\left[\left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2\right] = 2$$
- 4 GeV $< \sqrt{s} <$ 9 GeV, c quark contribute
$$R \to 2 + 3\left(\frac{2}{3}\right)^2 = \frac{10}{3}$$
- for $10 < \sqrt{s}$, b quark contribution
$$R = \frac{11}{3}$$



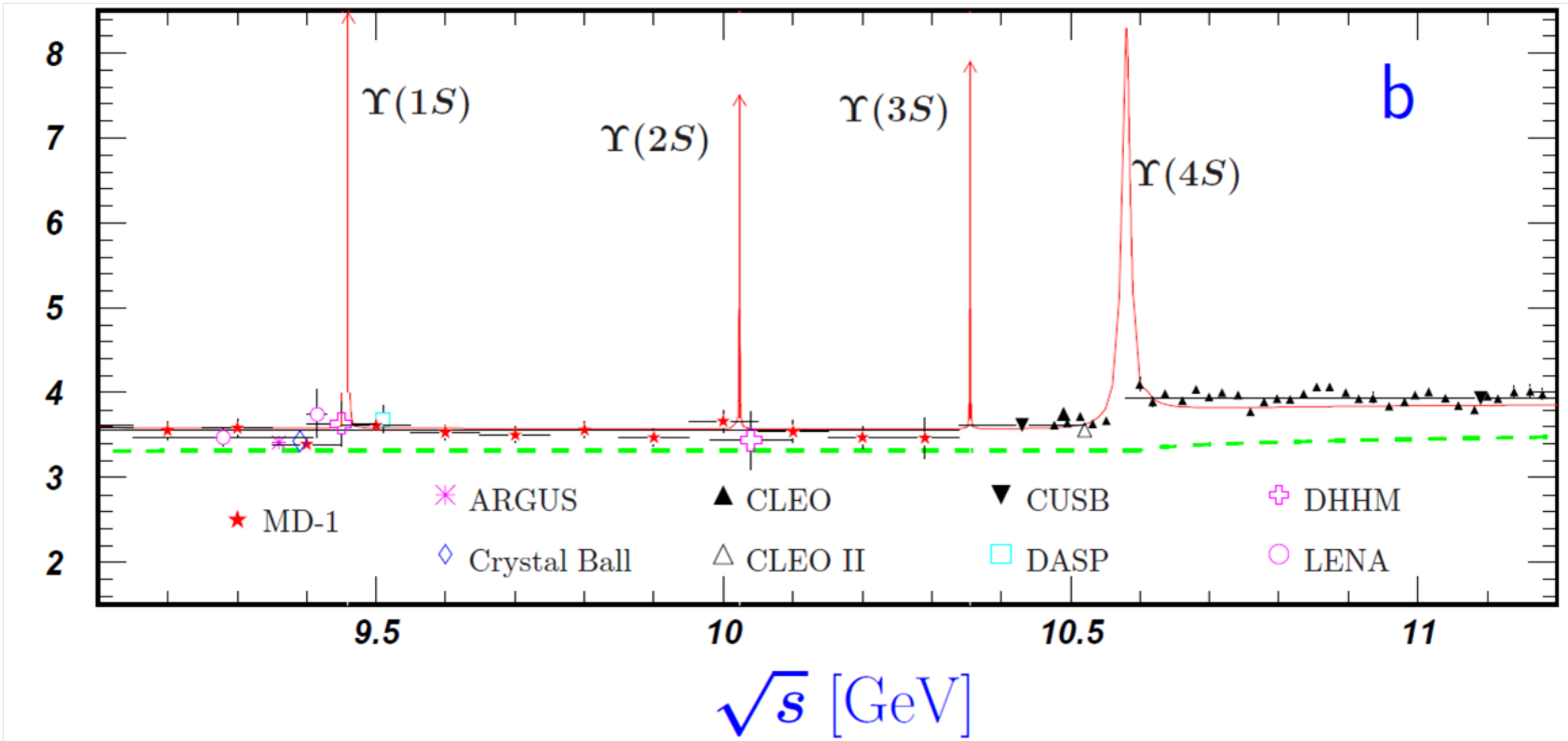
$$R \approx 2$$

Fig from: K. A. Olive et al. (Particle Data Group), Chin. Phys. C 38, 090001 (2014), and 2015 update. Reproduced from PDG.



$$R \approx \frac{10}{3}, \quad 4\ \text{GeV} < \sqrt{s} < 9\ \text{GeV}$$

Fig from: K. A. Olive et al. (Particle Data Group), Chin. Phys. C 38, 090001 (2014), and 2015 update. Reproduced from PDG.



$$R \approx \frac{11}{3}, \quad 10\,\mathrm{GeV} < \sqrt{s}$$

Fig from: K. A. Olive et al. (Particle Data Group), Chin. Phys. C 38, 090001 (2014), and 2015 update. Reproduced from PDG.

Decay of π^0

- $\pi^0 \to 2\gamma$
- $$\Gamma(\pi^0 \to 2\gamma) = N_c^2 (Q_u^2 - Q_d^2)^2 \frac{\alpha^2 m_{\pi^0}^3}{64\pi^3 F_\pi^2}$$

  $F_\pi = 91$ MeV is the pion decay constant, $Q_u = 2/3$, $Q_d = -1/3$
- $N_c = 3$

  $$\Gamma(\pi^0 \to 2\gamma) = 7.6\,\mathrm{eV}$$
- $$\Gamma(exp) = 7.80 \pm 0.12\,\mathrm{eV}$$

- Baryons

$$\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{10}$$

  singlet totally antisymmetric $\rightarrow \epsilon_{ijk} q^i q^j q^k$

- Mesons

$$\mathbf{3} \otimes \mathbf{3}^* = \mathbf{1} \oplus \mathbf{8}$$

  singlet $\rightarrow \frac{1}{\sqrt{3}} \delta_{ik} q^i q^{*k}$

- colour is not observed, hence we need singlet representation!
- the states

$$qq \rightarrow \mathbf{3} \otimes \mathbf{3} = \mathbf{3}^* \oplus \mathbf{6} \tag{2}$$

$$qqq \rightarrow \mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{3} \oplus \mathbf{3} \oplus \mathbf{3} \oplus \mathbf{6}^* \oplus \mathbf{6}^* \oplus \mathbf{15} \oplus \mathbf{15} \oplus \mathbf{15} \oplus \mathbf{15}' \tag{3}$$

$$\bar{q}\bar{q} \rightarrow \mathbf{3}^* \otimes \mathbf{3}^* = \mathbf{3} \oplus \mathbf{6}^* \tag{4}$$

$$\bar{q}\bar{q}\bar{q}\bar{q} \rightarrow \mathbf{3}^* \otimes \mathbf{3}^* \otimes \mathbf{3}^* \otimes \mathbf{3}^* = \mathbf{3}^* \oplus \mathbf{3}^* \oplus \mathbf{3}^* \oplus \mathbf{6} \oplus \mathbf{6} \oplus \mathbf{15}^* \oplus \mathbf{15}^* \oplus \mathbf{15}^* \oplus \mathbf{15}'^* \tag{5}$$

  are not allowed!

- **All hadron states and physical observables are colour singlets**

Why SU(3) for colour symmetry?

- We restrict ourselves to compact simple Lie groups
- We require compactness so that the unitary representations are finite dimensional.
- Additionally the group G for the colour symmetry should satisfy:
  1. the number of color degrees of freedom is three $\rightarrow$ quarks belong to the triplet representation
  2. antiquark state is different from the quark state, the triplet representation can not be real
  3. Mesons and baryons are singlets of the group G
  4. qq is not the singlet of the group G. The same condition holds also for other states such as $\bar{q}\bar{q}$, $\bar{q}\bar{q}\bar{q}\bar{q}$

- Lie groups are completely classified into the following classes:

$$A_N(=SU(N+1)), \quad B_N(SO(2N+1)), \quad C_N(=Sp(N)),$$

$$D_N(=SO(2N)),\ E_6\ E_7,\ E_8, F_4,\ G_2,$$

  where $N = 1, 2, 3, ...$ (for $D_N, N > 2$) and E_6, E_7, E_8, F_4 and G_2 are exceptional groups.
- Only $SU(2)$, $SU(3)$, $SO(3)$ and $Sp(1)$ have three dimensional irreducible representation
- $SU(2) \simeq Sp(1)$, $SO(3) \cong SU(2)/\mathbb{Z}_2$, and the triplet representations of these groups are real
- **Hence** $SU(3)$

Constructing the hadron wave function...

Resonance	Central Mass Value (MeV)	Total Width (MeV)	Elasticity	${}^{s}(I)_{J}$	$[SU(6), L^{P}]_{N}$	stars
$P_{33}(1232)$	1232	**120**	**1.00**	${}^{4}(10)_{3/2}$	$[56, 0^{+}]_{0}$	$\star\star\star\star$
$P_{11}(1440)$	1440	**350**	**0.65**	${}^{2}(8)_{1/2}$	$[56, 0^{+}]_{2}$	$\star\star\star\star$
$D_{13}(1520)$	1520	**120**	**0.55**	${}^{2}(8)_{3/2}$	$[70, 1^{-}]_{1}$	$\star\star\star\star$
$S_{11}(1535)$	1535	**150**	**0.40**	${}^{2}(8)_{1/2}$	$[70, 1^{-}]_{1}$	$\star\star\star\star$
$S_{31}(1620)$	1620	**150**	**0.25**	${}^{2}(10)_{1/2}$	$[70, 1^{-}]_{1}$	$\star\star\star\star$
$S_{11}(1650)$	1650	**150**	**0.67**	${}^{4}(8)_{1/2}$	$[70, 1^{-}]_{1}$	$\star\star\star\star$
$P_{33}(1600)$	1600	**350**	**0.20**	${}^{4}(10)_{3/2}$	$[56, 0^{+}]_{2}$	$\star\star\star$
$D_{13}(1700)$	1700	**100**	**0.10**	${}^{4}(8)_{3/2}$	$[70, 1^{-}]_{1}$	$\star\star\star$
$D_{15}(1675)$	1675	**150**	**0.55**	${}^{4}(8)_{5/2}$	$[70, 1^{-}]_{1}$	$\star\star\star\star$
$F_{15}(1680)$	1680	**130**	**0.65**	${}^{2}(8)_{5/2}$	$[56, 2^{+}]_{2}$	$\star\star\star\star$
$P_{11}(1710)$	1710	**100**	**0.15**	${}^{2}(8)_{1/2}$	$[70, 0^{+}]_{2}$	$\star\star\star$
$D_{33}(1700)$	1700	**300**	**0.15**	${}^{2}(10)_{3/2}$	$[70, 1^{-}]_{1}$	$\star\star\star$
$P_{13}(1720)$	1720	**150**	**0.15**	${}^{2}(8)_{3/2}$	$[56, 2^{+}]_{2}$	$\star\star\star\star$
$P_{31}(1910)$	1920	**250**	**0.22**	${}^{4}(10)_{1/2}$	$[56, 2^{+}]_{2}$	$\star\star\star\star$
$F_{35}(1905)$	1905	**350**	**0.10**	${}^{4}(10)_{5/2}$	$[56, 2^{+}]_{2}$	$\star\star\star\star$
$F_{37}(1950)$	1950	**300**	**0.37**	${}^{4}(10)_{7/2}$	$[56, 2^{+}]_{2}$	$\star\star\star\star$
$P_{33}(1920)$	1960	**200**	**0.12**	${}^{4}(10)_{3/2}$	$[56, 2^{+}]_{2}$	$\star\star\star$
$F_{17}(1990)$	1990	**349**	**0.05**	${}^{4}(8)_{7/2}$	$[70, 2^{+}]_{2}$	$\star\star$

Table: List of baryon resonances with hadronic invariant mass smaller than 2 GeV.

Hadron wave function:

$$\Psi_{hadron} = \Psi_{color} \times \Psi_{spin} \times \Psi_{flavor} \times \Psi_{space}. \tag{6}$$

- Space part is determined by the interactions between quarks – in the center of mass frame there is $O(3)$ symmetry for this part. Then space part can be described by the partial waves.
- Color part is given by $SU(3)$ singlet which is totally antisymmetric.
- Spin-flavour (assuming three quarks) should be given by representations:

$$SU_{spin}(2) \times SU_{flavor}(3) \subset SU(6). \tag{7}$$

- Given baryon wave function is characterized by total isospin and its third component, total spin (called later the angular momentum) J and its third component J_z.
- The space part, described by oscillator wave function, is characterized by two quantum numbers: N – the level in the oscillator, and angular momentum L and its third component L_z, where $\mathbf{J} = \mathbf{s} + \mathbf{L}$

Then the baryon wave function reads

$$|\mathcal{N}^*; J, J_z; I, I_z; N\rangle = \mathrm{Sym}\left\{ \sum_{s_z L_z, \rho\eta\kappa} \mathcal{C}^{\rho\eta\kappa}_{s_z L_z} \; | \, \mathcal{N}^*; I, I_z\rangle^{\rho} \; | \, \mathcal{N}^*; s, s_z\rangle^{\eta} \; | \, \mathcal{N}^*; N; L, L_z\rangle^{\kappa} \right\} \tag{8}$$

where: $J_z = s_z + L_z$; ρ, η, $\kappa = S, A, (M, S), (M, A)$ denote the symmetry types, namely: S (totally symmetric), A (totally antisymmetric), and M, S (mixed symmetric) M, A (mixed antisymmetric).

Nucleon Wave Function

$$qqq \quad \rightarrow \tag{9}$$

$$SU(2) \qquad \mathbf{2} \otimes \mathbf{2} \otimes \mathbf{2} = \quad \mathbf{2} \oplus \mathbf{2} \oplus \mathbf{4} \tag{10}$$

$$SU(3) \qquad \mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{10} \tag{11}$$

$$SU(6) \qquad \mathbf{6} \otimes \mathbf{6} \otimes \mathbf{6} = \mathbf{20} \oplus \mathbf{70} \oplus \mathbf{70} \oplus \mathbf{56} \tag{12}$$

Symmetric state $|\mathbf{56}\rangle_S$ may contain:

$$^4(10) \quad = \quad \left|\frac{3}{2}\right\rangle_S |10\rangle_S \tag{13}$$

$$^2(8) \quad = \quad \frac{1}{\sqrt{2}} \left(\left|\frac{1}{2}\right\rangle_{M,S} |8\rangle_{M,S} + \left|\frac{1}{2}\right\rangle_{M,A} |8\rangle_{M,A} \right) \tag{14}$$

Proton and neutron wave function $^2(8)$.

$$
\begin{aligned}
|P,8\rangle_{M,S} &= \frac{1}{\sqrt{6}}\left(|uud\rangle + |udu\rangle - 2\,|duu\rangle\right) \\
|P,8\rangle_{M,A} &= \frac{1}{\sqrt{2}}\left(|uud\rangle - |udu\rangle\right), \qquad (15) \\
|N,8\rangle_{M,S} &= -\frac{1}{\sqrt{6}}\left(|ddu\rangle + |dud\rangle - 2\,|udd\rangle\right), \qquad (16) \\
|N,8\rangle_{M,A} &= -\frac{1}{2}\left(2\cdot|ddu\rangle - 2\cdot|dud\rangle\right) \to -\frac{\sqrt{2}}{2}\left(|ddu\rangle - |dud\rangle\right). \\
&= -\frac{1}{\sqrt{2}}\left(|ddu\rangle - |dud\rangle\right). \qquad (17)
\end{aligned}
$$

Remark on conventions

Here the mixed-symmetry states are chosen (anti)symmetric in the **last two** quarks, while the tensor construction of Lecture 7 singled out the **first two**. Both bases span the same two-dimensional mixed subspace and are related by an orthogonal transformation.

In the case of the $\Delta(1232)$ $(\mathbf{10},4)$ resonance we get:

$$\left|\Delta^{++},10\right\rangle_S = |uuu\rangle_S = |uuu\rangle\,, \tag{18}$$

$$\left|\Delta^{+},10\right\rangle_S = |uud\rangle_S = \frac{1}{\sqrt{3}}\left(|uud\rangle + |udu\rangle + |duu\rangle\right), \tag{19}$$

$$\left|\Delta^{0},10\right\rangle_S = |udd\rangle_S = \frac{1}{\sqrt{3}}\left(|dud\rangle + |ddu\rangle + |udd\rangle\right), \tag{20}$$

$$\left|\Delta^{-},10\right\rangle_S = |ddd\rangle_S = |ddd\rangle\,. \tag{21}$$

For the spin $\frac{1}{2}$ there are two states of type M, S:

$$\left|\frac{1}{2},\frac{1}{2}\right\rangle_{M,S} = +|++-\rangle_{M,S} = \frac{1}{\sqrt{6}}\left(|++-\rangle + |+-+\rangle - 2\,|-++\rangle\right), \quad (22)$$

$$\left|\frac{1}{2},-\frac{1}{2}\right\rangle_{M,S} = -|--+\rangle_{M,S} = -\frac{1}{\sqrt{6}}\left(|--+\rangle + |-+-\rangle - 2\,|+--\rangle\right) \quad (23)$$

States of type M, A for spin $\frac{1}{2}$ read:

$$\left|\frac{1}{2},\frac{1}{2}\right\rangle_{M,A} = +|++-\rangle_{M,A} = \frac{1}{\sqrt{2}}\left(|++-\rangle - |+-+\rangle\right), \quad (24)$$

$$\left|\frac{1}{2},-\frac{1}{2}\right\rangle_{M,A} = -|--+\rangle_{M,A} = -\frac{1}{\sqrt{2}}\left(|--+\rangle - |-+-\rangle\right). \quad (25)$$

Magnetic moment of nucleon

Let us assume (Karl 1991):

(i) ground state is given by orbital S-wave, so only spin of quarks contributes to the magnetic moment

(ii) assume that magnetic moment of u-quark reads μ_u, while d-quark has magnetic moment μ_d and assume that $\mu_u = -2\mu_d$ and quark moments are proportional to their charges. On the other hand $\mu_d \neq \mu_s$.

From first assumption:

$$\hat{\mu}_z = \sum_i \mu_i \sigma_z^{(i)} \tag{26}$$

sum goes over three quarks!

Magnetic moment of the proton:

$$\mu(P) = \sum_i \langle P, s_z = \frac{1}{2} \mid \mu_i \sigma_3^{(i)} \mid P, s_z = \frac{1}{2} \rangle \tag{27}$$

$$= 3 \left\langle P, s_z = \frac{1}{2} \right| \mu_1 \sigma_3^{(1)} \left| P, s_z = \frac{1}{2} \right\rangle \tag{28}$$

$$= \frac{3}{2} \left(-\frac{1}{3} \cdot \frac{1}{3} \left(\mu_u + 2\mu_d \right) + \mu_u \right) \tag{29}$$

$$= \frac{3}{2} \left(\frac{8}{9} \mu_u - \frac{2}{9} \mu_d \right) \tag{30}$$

$$= \frac{1}{3} (4\mu_u - \mu_d) \tag{31}$$

Magnetic moment of neutron:

$$\mu(N) = \sum_i \left\langle N, s_z = \frac{1}{2} \middle| \mu_i \sigma_3^{(i)} \middle| N, s_z = \frac{1}{2} \right\rangle \tag{32}$$

$$= 3 \left\langle N, s_z = \frac{1}{2} \middle| \mu_1 \sigma_3^{(1)} \middle| N, s_z = \frac{1}{2} \right\rangle \tag{33}$$

$$= \frac{3}{2} \left(-\frac{1}{3} \cdot \frac{1}{3} (\mu_d + 2\mu_u) + \mu_d \right) \tag{34}$$

$$= \frac{3}{2} \left(\frac{8}{9} \mu_d - \frac{2}{9} \mu_u \right) \tag{35}$$

$$= \frac{1}{3} (4\mu_d - \mu_u) \tag{36}$$

Then using the second assumption:

$$\mu(P) = \frac{3\mu_u}{2}, \quad \mu(N) = -\mu_u \tag{37}$$

Hence:

$$\frac{\mu(P)}{\mu(N)} = -\frac{3}{2} = -1.5 \tag{38}$$

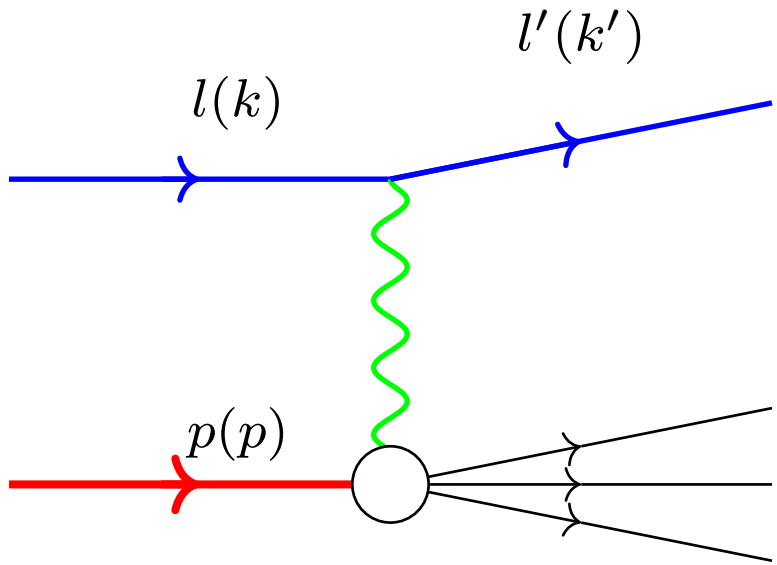
Experimental value ≈ -1.46!

Deep Inelastic Scattering (DIS)

Let us consider the inelastic lepton scattering off the nucleon (proton),

$$l(k) + p(p) \to l'(k') + X(p_1 + p_2 + ...). \tag{39}$$

X denotes any configuration of hadrons in the final state, while p_1, p_2, ... are the four-momenta of the hadrons in the final state.



$$k^\mu = (E, \mathbf{k}), \quad k'^\mu = (E', \mathbf{k}'), \quad q \equiv k - k', \quad q^2 = -Q^2$$

M – proton mass, θ is the scattering angle (between $\mathbf{k}$ and $\mathbf{k}'$)

The cross-section (in the Lab frame) reads

$$d\sigma_{LAB} = \frac{1}{4EM} \underbrace{\sum_{s,s'} \frac{1}{2} \frac{d^3\mathbf{k}'}{(2\pi)^3 2E'}}_{lepton} \sum_X \prod_{i=1}^{X} \frac{d^3\mathbf{p_i}}{(2\pi)^3 2E_{p_i}}$$
$$\frac{1}{2} \sum_{spins} |\mathcal{M}(k, k', s, s'; X, spins)|^2 (2\pi)^4 \delta^4 \left(p + k - k' - p_X\right) \quad (40)$$

where

$$p_X = p_1 + p_2 + ... \quad (41)$$

The lepton part of (40) can be easily simplified

$$d^3\mathbf{k}' = |\mathbf{k}'|^2 d\Omega d|\mathbf{k}'| = |\mathbf{k}'|E' d\Omega dE'$$

$$\frac{d\sigma}{d\Omega dE'}_{LAB} = \frac{|\mathbf{k}'|}{8EM}\sum_{s,s'}\frac{1}{2(2\pi)^3}\sum_X\int\prod_{i=1}^{X}\frac{d^3\mathbf{p_i}}{(2\pi)^3 2E_{p_i}}$$
$$\frac{1}{2}\sum_{spins}|\mathcal{M}(k,k',s,s';X,spins)|^2(2\pi)^4\delta^4\left(p+k-k'-p_X\right) \quad (42)$$

Electron-proton scattering

Let us consider (deep)-inelastic electron scattering off the proton, then

$$e + p \to e + X \tag{43}$$

In the case of the one particular final hadronic state of the four-momentum $\mathbf{p}'$ the matrix element has simple form,

$$i\mathcal{M}_{fi}^{(1)} = -\frac{ie^2}{q^2}\overline{u}(\mathbf{k}', s')\gamma_\mu u(\mathbf{k}, s)\langle \mathbf{p}', r' \mid \mathcal{J}_{had}^{\mu}(0) \mid \mathbf{p}, r\rangle. \tag{44}$$

For any final hadronic state we shall sum over all possible multi-particle states

$$\frac{d\sigma}{d\Omega dE'}_{LAB} = \frac{e^4|\mathbf{k}'|}{32\pi^3 q^4 E}\frac{1}{2}\sum_{s,s'}\overline{u}(\mathbf{k}',s')\gamma_\mu u(\mathbf{k},s)(\overline{u}(\mathbf{k}',s')\gamma_\nu u(\mathbf{k},s))^*$$

$$\frac{1}{2M}\sum_X\sum_{spins}\int\prod_{i=1}^{X}\frac{d^3\mathbf{p_i}}{(2\pi)^3 2E_{p_i}}\frac{1}{2}(2\pi)^4\delta^4\left(p+k-k'-p_X\right)$$

$$\langle\mathbf{p}_X \mid \mathcal{J}^\mu_{had}(0)\mid \mathbf{p},r\rangle\langle\mathbf{p}_X \mid \mathcal{J}^\nu_{had}(0)\mid \mathbf{p},r\rangle^* \tag{45}$$

where $|\mathbf{p}_X\rangle$ is properly normalized hadronic multi-particle state,

$$|\mathbf{p}_X\rangle \sim |\mathbf{p}_1\rangle\otimes|\mathbf{p}_2\rangle\otimes\ldots \tag{46}$$

$$\frac{d\sigma}{d\Omega dE'}_{LAB} = \frac{\alpha^2|\mathbf{k}'|}{2\pi q^4 E}\frac{1}{2}\sum_{s,s'}\overline{u}(\mathbf{k}',s')\gamma_\mu u(\mathbf{k},s)\overline{u}(\mathbf{k},s)\gamma_\nu u(\mathbf{k}',s')$$

$$\frac{1}{4M}\sum_X\sum_{spins}\int\prod_{i=1}^{X}\frac{d^3\mathbf{p_i}}{(2\pi)^3 2E_{p_i}}(2\pi)^4\delta^4\left(p+k-k'-p_X\right)$$

$$\langle\mathbf{p}_X\mid\mathcal{J}^\mu_{had}(0)\mid\mathbf{p},r\rangle\langle\mathbf{p}_X\mid\mathcal{J}^\nu_{had}(0)\mid\mathbf{p},r\rangle^*$$

$$= \frac{\alpha^2|\mathbf{k}'|}{q^4 E}\frac{1}{2}\sum_{s,s'}\overline{u}(\mathbf{k}',s')\gamma_\mu u(\mathbf{k},s)\overline{u}(\mathbf{k},s)\gamma_\nu u(\mathbf{k}',s')$$

$$\frac{1}{4M}\sum_X\sum_{spins}\int\prod_{i=1}^{X}\frac{d^3\mathbf{p_i}}{(2\pi)^3 2E_{p_i}}(2\pi)^3\delta^4\left(p+k-k'-p_X\right)$$

$$\langle\mathbf{p}_X\mid\mathcal{J}^\mu_{had}(0)\mid\mathbf{p},r\rangle\langle\mathbf{p}_X\mid\mathcal{J}^\nu_{had}(0)\mid\mathbf{p},r\rangle^* \qquad (47)$$

Let us define the leptonic and hadronic tensor,

$$L^{\mu\nu} = \frac{1}{2}\sum_{s,s'}\overline{u}(\mathbf{k}',s')\gamma^{\mu}u(\mathbf{k},s)\overline{u}(\mathbf{k},s)\gamma^{\nu}u(\mathbf{k}',s') \tag{48}$$

$$= \frac{1}{2}\mathrm{Tr}\hat{k}'\gamma^{\mu}\hat{k}\gamma^{\nu} = 2(k^{\mu}k'^{\nu} + k'^{\mu}k^{\nu} - k\cdot k' g^{\mu\nu}), \tag{49}$$

$$W^{\mu\nu} = \frac{1}{4M}\sum_{X}\sum_{spins}\int\prod_{i=1}^{X}\frac{d^3\mathbf{p_i}}{(2\pi)^3 2E_{p_i}}(2\pi)^3\delta^4\left(p+k-k'-p_X\right)$$
$$\langle\mathbf{p},r\mid\mathcal{J}^{\nu\dagger}_{had}(0)\mid\mathbf{p}_X\rangle\langle\mathbf{p}_X\mid\mathcal{J}^{\mu}_{had}(0)\mid\mathbf{p},r\rangle \tag{50}$$

Then the differential cross section reads,

$$\frac{d\sigma}{d\Omega dE'}_{LAB} = \frac{\alpha^2}{q^4}\frac{|\mathbf{k}'|}{E}L^{\mu\nu}W_{\mu\nu} = \frac{\alpha^2}{q^4}\frac{E'}{E}L^{\mu\nu}W_{\mu\nu} \tag{51}$$

We shall use two relations,

$$\mathbf{1} = \sum_X \mid \mathbf{p}_X\rangle\langle \mathbf{p}_X \mid = \sum_X \int \prod_{i=1}^{X} \frac{d^3 p_i}{(2\pi)^3\, 2E_{p_i}} \mid \mathbf{p}_i\rangle\langle \mathbf{p}_i \mid \tag{52}$$

and

$$\mathcal{J}(y) = e^{i\mathcal{P}\cdot y}\mathcal{J}(0)e^{-i\mathcal{P}\cdot y}, \quad \mathcal{J}^\dagger(y) = e^{i\mathcal{P}\cdot y}\mathcal{J}^\dagger(0)e^{-i\mathcal{P}\cdot y}. \tag{53}$$

Hence

$$\langle \mathbf{p}, r \mid \mathcal{J}^{\mu\dagger}_{had}(y) \mid \mathbf{p}_X\rangle = \langle \mathbf{p}, r \mid \mathcal{J}^{\mu\dagger}_{had}(0) \mid \mathbf{p}_X\rangle e^{iy\cdot(p-p_X)}. \tag{54}$$

Then the hadronic tensor can be written as,

$$W^{\mu\nu} = \frac{1}{4M}\sum_X \sum_{spins} \int \prod_{i=1}^{X} \frac{d^3\mathbf{p_i}}{(2\pi)^3 2E_{p_i}} (2\pi)^3 \delta^4 (p+q-p_X) \times \langle \mathbf{p}, r \mid \mathcal{J}^{\nu\dagger}_{had}(0) \mid \mathbf{p}_X \rangle \langle \mathbf{p}_X \mid \mathcal{J}^{\mu}_{had}(0) \mid \mathbf{p}, r \rangle \quad (55)$$

where we use the property

$$(2\pi)^4 \delta^4 (p+q-p_X) = \int d^4 y e^{iy(p+q-p_X)} \quad (56)$$

$$W^{\mu\nu} = \frac{1}{4M}\frac{1}{2\pi} \int d^4 y \sum_X \sum_{spins} \int \prod_{i=1}^{X} \frac{d^3\mathbf{p_i}}{(2\pi)^3 2E_{p_i}} e^{iy\cdot(q+p-p_X)} \times \langle \mathbf{p}, r \mid \mathcal{J}^{\nu\dagger}_{had}(0) \mid \mathbf{p}_X \rangle \langle \mathbf{p}_X \mid \mathcal{J}^{\mu}_{had}(0) \mid \mathbf{p}, r \rangle \quad (57)$$

$$
\begin{aligned}
W^{\mu\nu} &= \frac{1}{4M}\frac{1}{2\pi}\int d^4y \sum_X \sum_{spins} \int \prod_{i=1}^{X} \frac{d^3\mathbf{p_i}}{(2\pi)^3 2E_{p_i}} e^{iy\cdot(q+p-p_X)} \\
&\times \langle \mathbf{p}, r \mid \mathcal{J}^{\nu\dagger}_{had}(y) \mid \mathbf{p}_X\rangle\langle \mathbf{p}_X \mid \mathcal{J}^{\mu}_{had}(0) \mid \mathbf{p}, r\rangle e^{-iy\cdot(p-p_X)} \\
&= \frac{1}{4M}\frac{1}{2\pi}\int d^4y \langle \mathbf{p}, r \mid \mathcal{J}^{\nu\dagger}_{had}(y)\mathcal{J}^{\mu}_{had}(0) \mid \mathbf{p}, r\rangle e^{iy\cdot q} \qquad (58)
\end{aligned}
$$

Now notice that

$$
\begin{aligned}
* &= \int d^4y \langle \mathbf{p} \mid \mathcal{J}^{\mu}_{had}(0)\mathcal{J}^{\nu\dagger}_{had}(y) \mid \mathbf{p}\rangle e^{iy\cdot q} \\
&= \sum_X \sum_{spins} \int \prod_{i=1}^{X} \frac{d^3\mathbf{p_i}}{(2\pi)^3 2E_{p_i}} \int d^4y \langle \mathbf{p} \mid \mathcal{J}^{\mu}_{had}(0) \mid \mathbf{p}_X\rangle\langle \mathbf{p}_X \mid \mathcal{J}^{\nu\dagger}_{had}(y) \mid \mathbf{p}\rangle e^{iy\cdot q} \\
&= \sum_X \sum_{spins} \int \prod_{i=1}^{X} \frac{d^3\mathbf{p_i}}{(2\pi)^3 2E_{p_i}} \int d^4y \times \\
&\quad \langle \mathbf{p} \mid \mathcal{J}^{\mu}_{had}(0) \mid \mathbf{p}_X\rangle\langle \mathbf{p}_X \mid \mathcal{J}^{\nu\dagger}_{had}(0) \mid \mathbf{p}\rangle e^{iy\cdot q + iy\cdot(p_X - p)} \\
&= \sum_X \sum_{spins} \int \prod_{i=1}^{X} \frac{d^3\mathbf{p_i}}{(2\pi)^3 2E_{p_i}} \\
\times &\quad \langle \mathbf{p} \mid \mathcal{J}^{\mu}_{had}(0) \mid \mathbf{p}_X\rangle\langle \mathbf{p}_X \mid \mathcal{J}^{\nu\dagger}_{had}(0) \mid \mathbf{p}\rangle (2\pi)^4 \delta^{(4)}(q + p_X - p) = 0 \qquad (59)
\end{aligned}
$$

- The delta function requires $E_X = M - \nu < M$, which is impossible for a physical hadronic intermediate state with baryon number one.

Hence we may write the final formula,

$$W^{\mu\nu} = \frac{1}{4M}\frac{1}{2\pi}\int d^4y \langle \mathbf{p}, r \mid \left[\mathcal{J}^{\nu\dagger}_{had}(y), \mathcal{J}^{\mu}_{had}(0)\right] \mid \mathbf{p}, r\rangle e^{iy\cdot q}$$

From the current conservation we see immediately that,

$$\begin{aligned} q_\nu W^{\mu\nu} &= \frac{1}{4M}\frac{1}{2\pi}\int d^4y \langle \mathbf{p}, r \mid q_\nu \left[\mathcal{J}^{\nu\dagger}_{had}(y), \mathcal{J}^{\mu}_{had}(0)\right] \mid \mathbf{p}, r\rangle e^{iy\cdot q} \\ &= \frac{1}{4M}\frac{1}{2\pi}\int d^4y \langle \mathbf{p}, r \mid \left[\mathcal{J}^{\nu\dagger}_{had}(y), \mathcal{J}^{\mu}_{had}(0)\right] \mid \mathbf{p}, r\rangle \frac{1}{i}\partial_\nu e^{iy\cdot q} \\ &= -\frac{1}{4M}\frac{1}{2\pi i}\int d^4y \langle \mathbf{p}, r \mid \left[\partial_\nu \mathcal{J}^{\nu\dagger}_{had}(y), \mathcal{J}^{\mu}_{had}(0)\right] \mid \mathbf{p}, r\rangle e^{iy\cdot q} = 0. \end{aligned}$$

It is based on the constraint coming from gauge invariance

$$\partial_\mu \mathcal{J}^{\mu}_{had} = 0$$

- Since the spin-averaged leptonic tensor is symmetric, only the symmetric part of the hadronic tensor contributes to the unpolarized electromagnetic cross section.
- We have only p and q four-momenta in the game, hence, there are only two independent Lorenz scalars: q^2, $p \cdot q$
- The most general form of the Lorentz invariant hadronic tensor reads

$$\begin{aligned} W^{\mu\nu} &= -g^{\mu\nu} W_1(q^2, p\cdot q) + \frac{p^\mu p^\nu}{M^2} W_2(q^2, p\cdot q) + \frac{q^\mu q^\nu}{M^2} W_4(q^2, p\cdot q) \\ &\quad + (p^\mu q^\nu + p^\nu q^\mu)\, \frac{W_5(q^2, p\cdot q)}{M^2}. \end{aligned} \tag{60}$$

The gauge invariance (charge conservation) constraints the tensor as follows

$$q_\mu W^{\mu\nu} = 0 \Rightarrow q^2 W_5 = -W_2\, p\cdot q, \quad q^2 W_4 = -W_5\, p\cdot q + W_1 M^2. \tag{61}$$

Then

$$q_\mu W^{\mu\nu} = 0 \Rightarrow q^2 W_5 = -W_2 p\cdot q, \quad q^2 W_4 = W_2 \frac{(p\cdot q)^2}{q^2} + W_1 M^2. \tag{62}$$

Then the gauge invariant tensor reads

$$W_{\mu\nu} = \left(p_\mu - q_\mu \frac{p\cdot q}{q^2}\right)\left(p_\nu - q_\nu \frac{p\cdot q}{q^2}\right)\frac{W_2}{M^2} + \left(\frac{q_\mu q_\nu}{q^2} - g_{\mu\nu}\right) W_1 \tag{63}$$

We easily compute the differential cross section,

$$\frac{d^2\sigma}{d\Omega dE'}_{LAB} = \frac{\alpha^2}{4E^2 \sin^4\frac{\theta}{2}}\left(2W_1 \sin^2\frac{\theta}{2} + W_2 \cos^2\frac{\theta}{2}\right) \tag{64}$$

Dimension:

$$Left : length^3$$

Hence $W_{1,2} = length.$

Structure Functions

Let us introduce F_1 and F_2 (dimensionless) structure functions

$$F_1 = MW_1 \quad F_2 = \nu W_2. \tag{65}$$

- For large q^2 transfer DIS of electron with proton can be modelled as an interaction of electron with point-like constituents inside the nucleon, called partons.
- Define dimensionless scaling variable x

$$x = -\frac{q^2}{2M\nu} = \frac{Q^2}{2M\nu}$$

  where $0 \leq x \leq 1$, for $x = 1$ elastic scattering, while for $x \to 0$ deep inelastic scattering.
- $$y = \frac{\nu}{E} = 1 - \frac{E'}{E}$$
- $$dxdy = \frac{E'}{E}\frac{d\Omega dE'}{2\pi y M}$$
- then

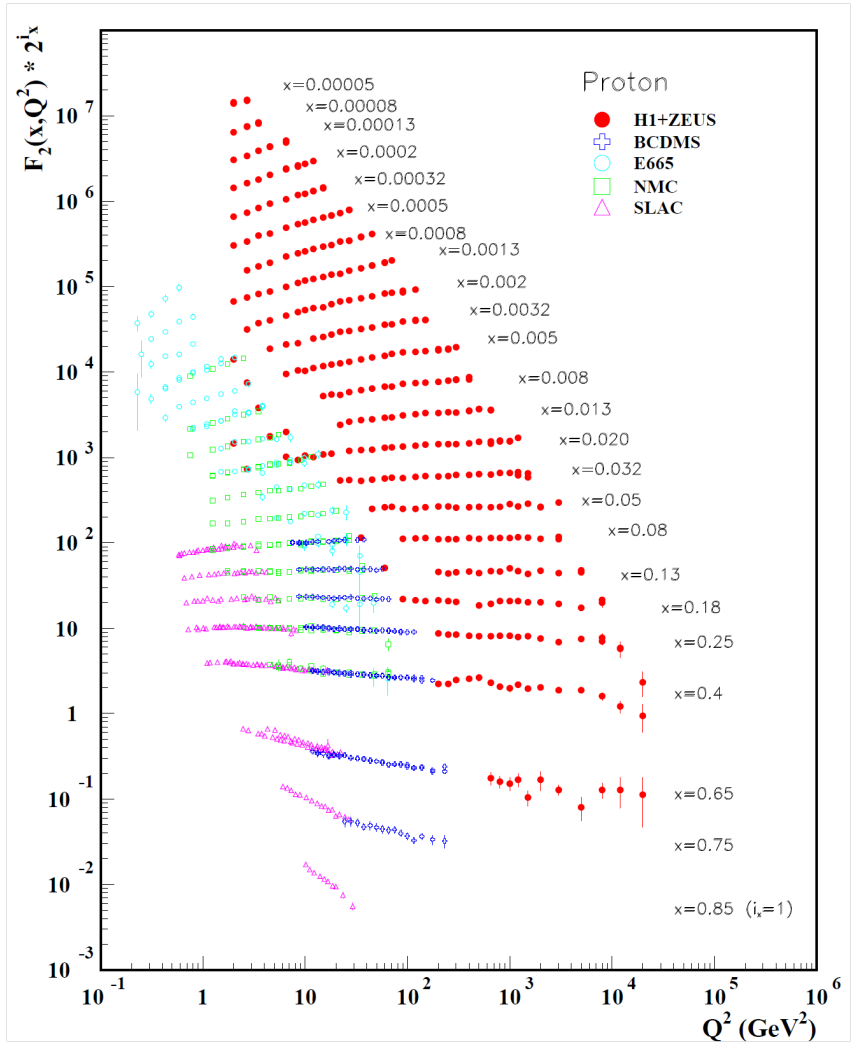
$$\frac{d^2\sigma}{dxdy} = \frac{4\pi\alpha^2}{xyQ^2}\left[xy^2F_1 + \left(1 - y - \frac{M}{2E}xy\right)F_2\right] \tag{66}$$

Bjorken scaling

Bjorken scaling

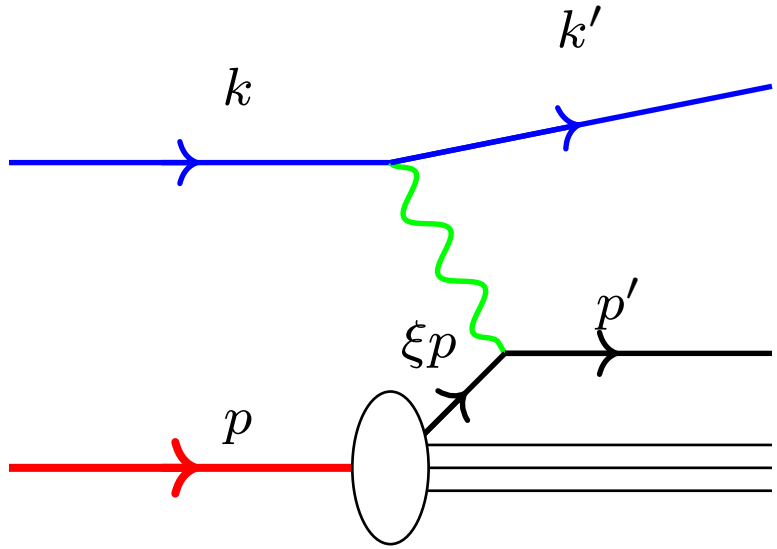
In the large Q^2 limit with x fixed, the F_is are functions x only

$$\lim_{Q^2 \to \infty} F_i \left(x, \frac{q^2}{M^2} \right) \to F_i(x)$$



Source: K. A. Olive et al. (Particle Data Group), Chin. Phys. C 38, 090001 (2014), and 2015 update. Reproduced from PDG.

Parton Model



- In the Bjorken limit, we neglect the nucleon mass in the parton kinematics and approximate the parton four-momentum by ξp.
- The parton is treated as massless: $(\xi p)^2 \simeq 0$.

$$W^{\mu\nu} \quad \rightarrow \quad K^{\mu\nu} \tag{67}$$

$$K^{\mu\nu} \quad = \quad \frac{1}{4\xi M} \sum_{spins} \int \frac{d^3\mathbf{p}'}{(2\pi)^3 2E_{p'}} (2\pi)^3 \delta^{(4)} \left(\xi p + k - k' - p'\right) \tag{68}$$

$$
\begin{aligned}
K^{\mu\nu} &= \frac{1}{4\xi M}\sum_{spins}\int\frac{d^3\mathbf{p}'}{(2\pi)^3 2E_{p'}}(2\pi)^3\delta^{(4)}\left(\xi p + k - k' - p'\right) \\
&\quad \langle \mathbf{p}' \mid \mathcal{J}^{\mu}_{had}(0) \mid \xi\mathbf{p}, r\rangle\langle \mathbf{p}' \mid \mathcal{J}^{\nu}_{had}(0) \mid \xi\mathbf{p}, r\rangle^* \\
&= \frac{1}{4\xi M}\sum_{spins}\frac{\delta\left(\xi p_0 + q_0 - {p'}_0\right)}{2{p'}_0}\langle \mathbf{p}' \mid \mathcal{J}^{\mu}_{had}(0) \mid \xi\mathbf{p}, r\rangle\langle \mathbf{p}' \mid \mathcal{J}^{\nu}_{had}(0) \mid \xi\mathbf{p}, r\rangle^* \\
&= \frac{1}{4\xi M}\sum_{spins}\frac{\delta\left(\xi p_0 + q_0 - {p'}_0\right)}{2{p'}_0}\underbrace{\bar{u}(\xi\mathbf{p})\gamma^{\mu}u(\mathbf{p}')\bar{u}(\mathbf{p}')\gamma^{\nu}u(\xi\mathbf{p})}_{*} \qquad (69)
\end{aligned}
$$

- We neglect the parton masses:

$$
\begin{aligned}
* &= \sum_{spin} \bar{u}(\xi\mathbf{p})\gamma_\mu u(\mathbf{p}')\bar{u}(\mathbf{p}')\gamma_\nu u(\xi\mathbf{p}) && (70)\\
&= \mathrm{Tr}\,\hat{p}'\gamma_\nu\hat{\xi p}\gamma_\mu && (71)\\
&= 4\left(p'_\nu \xi p_\mu + p'_\mu \xi p_\nu - g_{\mu\nu} p'\cdot(\xi p)\right) && (72)\\
&= 4\left(2\xi^2 p_\mu p_\nu + q_\mu \xi p_\nu + q_\nu \xi p_\mu - g_{\mu\nu}((\xi p)^2 + \xi\, q\cdot p)\right) && (73)\\
&= 8\xi^2 p_\mu p_\nu + 4\xi(q_\mu p_\nu + q_\nu p_\mu) - 4g_{\mu\nu}((\xi p)^2 + \xi q\cdot p) && (74)\\
&= 8\xi^2 p_\mu p_\nu + 4\xi(q_\mu p_\nu + q_\nu p_\mu) - 4g_{\mu\nu}((\xi p)^2 + \xi M\nu) && (75)
\end{aligned}
$$

- Notice that $\xi p\cdot q = \xi M\nu$ and $(\xi p)^2 = p'^2 = 0$
- Notice that

$$L_{\mu\nu}q^\mu = L_{\mu\nu}q^\nu = 0.$$

- Therefore, upon contraction with the leptonic tensor, we can use

$$* \doteq 8\xi^2 p_\mu p_\nu - 4g_{\mu\nu}\xi M\nu. \qquad (76)$$

$$* = \frac{\delta(p_0' - \xi p_0 - q_0)}{2p_0'} = \theta(\xi p_0 + q_0)\delta\left(p_0'^2 - (\xi p_0 + q_0)^2\right). \tag{77}$$

but $\mathbf{p}' = \xi\mathbf{p} + \mathbf{q}$ hence

$$* = \theta(\xi p_0 + q_0)\delta(p'^2 - (\xi p + q)^2) \tag{78}$$

In the Bjorken limit, neglecting target-mass corrections, we take $(\xi p)^2 \simeq 0$.

$$* = \theta(\xi p_0 + q_0)\delta((p')^2 - (\xi p)^2 - q^2 - 2M\nu\xi) \to \theta(p_0')\delta(q^2 + 2M\nu\xi) \tag{79}$$

$$= \theta(\xi p_0 + q_0)\frac{\delta(\xi - x)}{2M\nu} \tag{80}$$

$$x = -\frac{q^2}{2M\nu}$$

$$x_B = \frac{Q^2}{2M\nu}, \qquad Q^2 = -q^2.$$

Thus, the parton momentum fraction is identified with Bjorken x:

$$\boxed{\xi = x_B}.$$

Callan-Gross relation

- $$K_{\mu\nu} = \frac{1}{\xi M}\theta(\xi p_0 + q_0)\frac{\delta(\xi - x)}{2M\nu}\left(2\xi^2 p_\mu p_\nu - g_{\mu\nu}\xi M\nu\right) \quad (81)$$

$$= \theta(\xi p_0 + q_0)\delta(\xi - x)\left(\frac{\xi}{\nu}\frac{p_\mu p_\nu}{M^2} - \frac{1}{2M}g_{\mu\nu}\right) \quad (82)$$

- Let $f(\xi)$ be the number of partons with momenta between ξ and $\xi + d\xi$ Then the hadronic tensor reads

$$W_{\mu\nu} = \int_0^1 f(\xi) K_{\mu\nu} d\xi \quad (83)$$

- On the other hand

$$W_{\mu\nu} = \frac{p_\mu p_\nu}{M^2} W_2 - g_{\mu\nu} W_1 \quad (84)$$

hence

$$W_1 = \frac{f(x)}{2M}, \quad W_2 = \frac{x f(x)}{\nu} \quad (85)$$

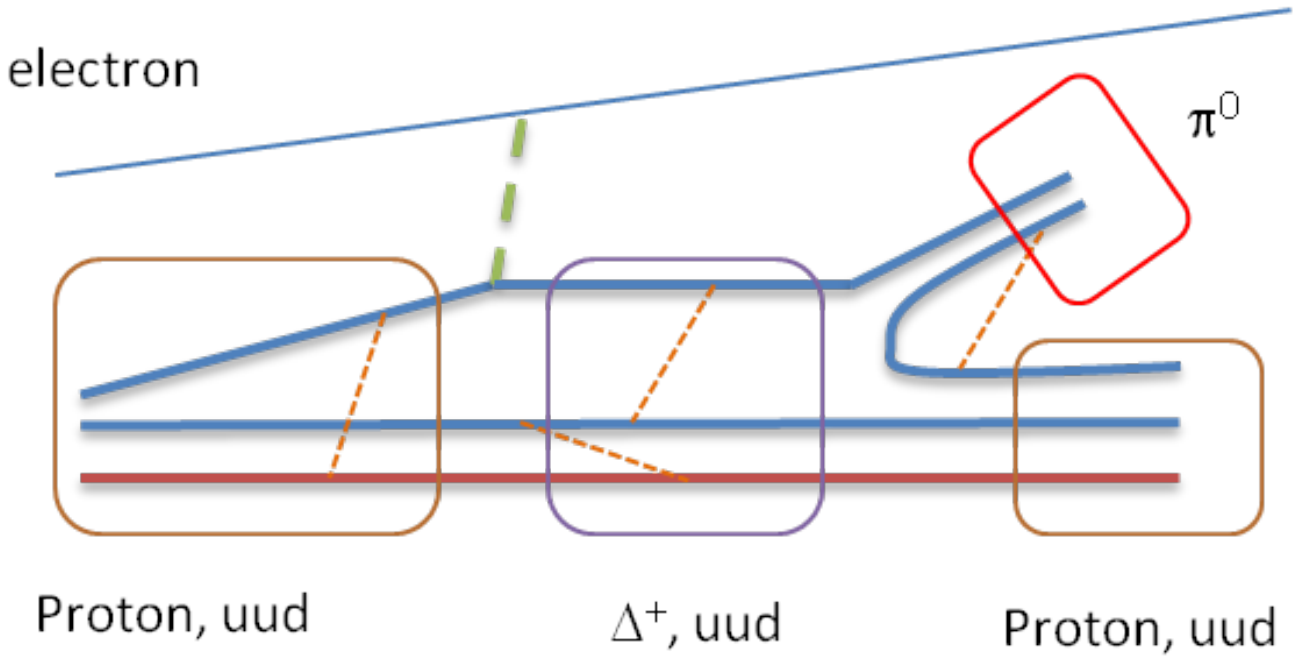
so we get

$$F_1 = \frac{1}{2} f(x), \quad F_2 = x f(x) \quad (86)$$

- $\rightarrow$ **Callan-Gross relation**

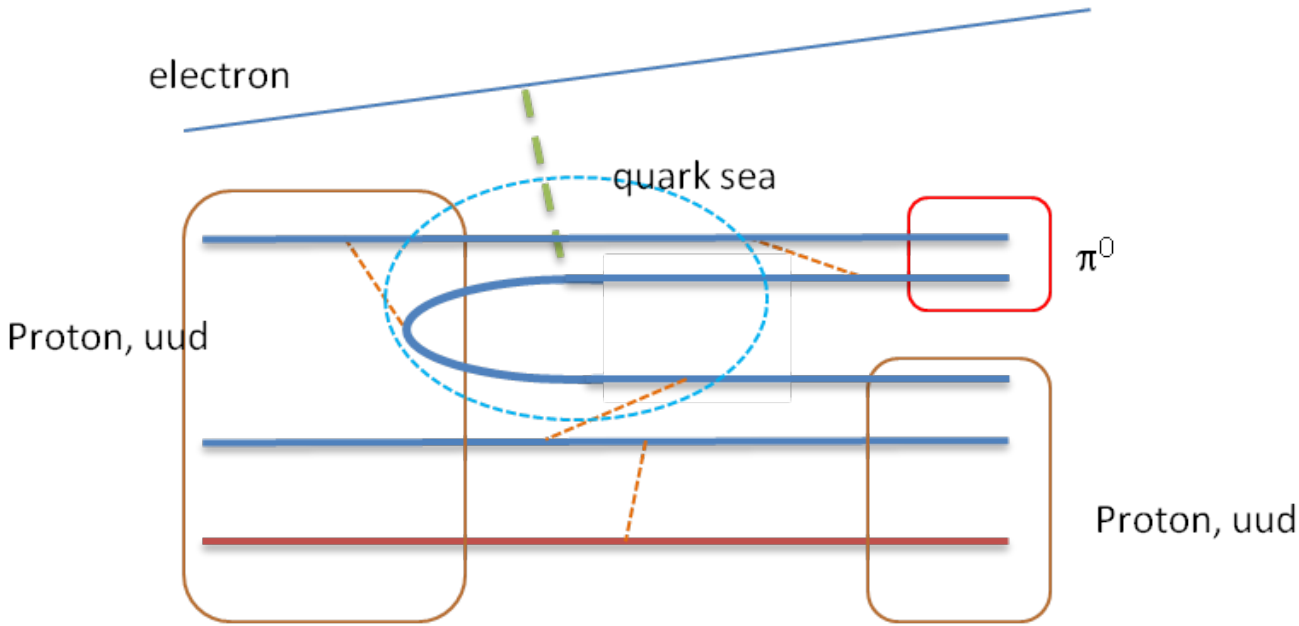
$$2x F_1(x) = F_2(x) \quad (87)$$

it is a direct consequence of the spin-$1/2$ nature of partons, indeed for spinless particles $F_1(x) = 0$



Remark!

Hadronic Intermediate State, $\Delta(1232)$ and others...



Remark!

Non-resonant mechanism

Parton Distribution Functions (PDFs)

- the quark electromagnetic current has the form

$$J^{\mu}_{em} = \frac{2}{3}\bar{u}(x)\gamma^{\mu}u(x) - \frac{1}{3}\bar{d}(x)\gamma^{\mu}d(x) - \frac{1}{3}\bar{s}(x)\gamma^{\mu}s(x) \tag{88}$$

then it is easy to see that

$$\begin{aligned} 2F_1^{ep}(x) &= \frac{4}{9}(u_p + \bar{u}_p) + \frac{1}{9}(d_p + \bar{d}_p) + \frac{1}{9}(s_p + \bar{s}_p) \\ 2F_1^{en}(x) &= \frac{4}{9}(u_n + \bar{u}_n) + \frac{1}{9}(d_n + \bar{d}_n) + \frac{1}{9}(s_n + \bar{s}_n) \end{aligned}$$

where $q_N(x)$ denotes the probability of finding a parton with longitudinal momentum fraction x carrying the quantum number of nucleon.

Parton distribution functions are constrained by quantum number of nucleon

- Isospin (the third component of the proton isospin $= 1/2$)

$$\int_0^1 \left[\frac{1}{2}(u_p(x) - \bar{u}_p(x)) - \frac{1}{2}(d_p(x) - \bar{d}_p(x))\right] dx = \frac{1}{2} \tag{89}$$

- Strangeness

$$\int_0^1 (s_p(x) - \bar{s}_p(x))dx = 0 \tag{90}$$

- Proton charge $(+1)$

$$\int_0^1 \left[\frac{2}{3}(u_p(x) - \bar{u}_p(x)) - \frac{1}{3}(d_p(x) - \bar{d}_p(x)) - \frac{1}{3}(s_p(x) - \bar{s}_p(x))\right] dx = 1 \tag{91}$$

Isospin symmetry: $u \leftrightarrow d$

$$u_p(x) = d_n(x) \equiv u(x) \quad (92)$$
$$d_p(x) = u_n(x) \equiv d(x) \quad (93)$$
$$s_p(x) = s_n(x) \equiv s(x) \quad (94)$$

In this case

$$2F_1^{ep}(x) = \frac{4}{9}(u+\bar{u}) + \frac{1}{9}(d+\bar{d}) + \frac{1}{9}(s+\bar{s})$$
$$2F_1^{en}(x) = \frac{4}{9}(d+\bar{d}) + \frac{1}{9}(u+\bar{u}) + \frac{1}{9}(s+\bar{s})$$

Valence-Sea

$$q(x) = q_{valence}(x) + q_{sea}(x) \tag{95}$$

- Presence of valence quarks supported by quark model
- Sea quark pairs produced by the gluons
- For illustration, for the proton, assume the additional simplified ansatz $u_v(x) = 2d_v(x)$. This is not a consequence of valence-quark counting. Additionally,

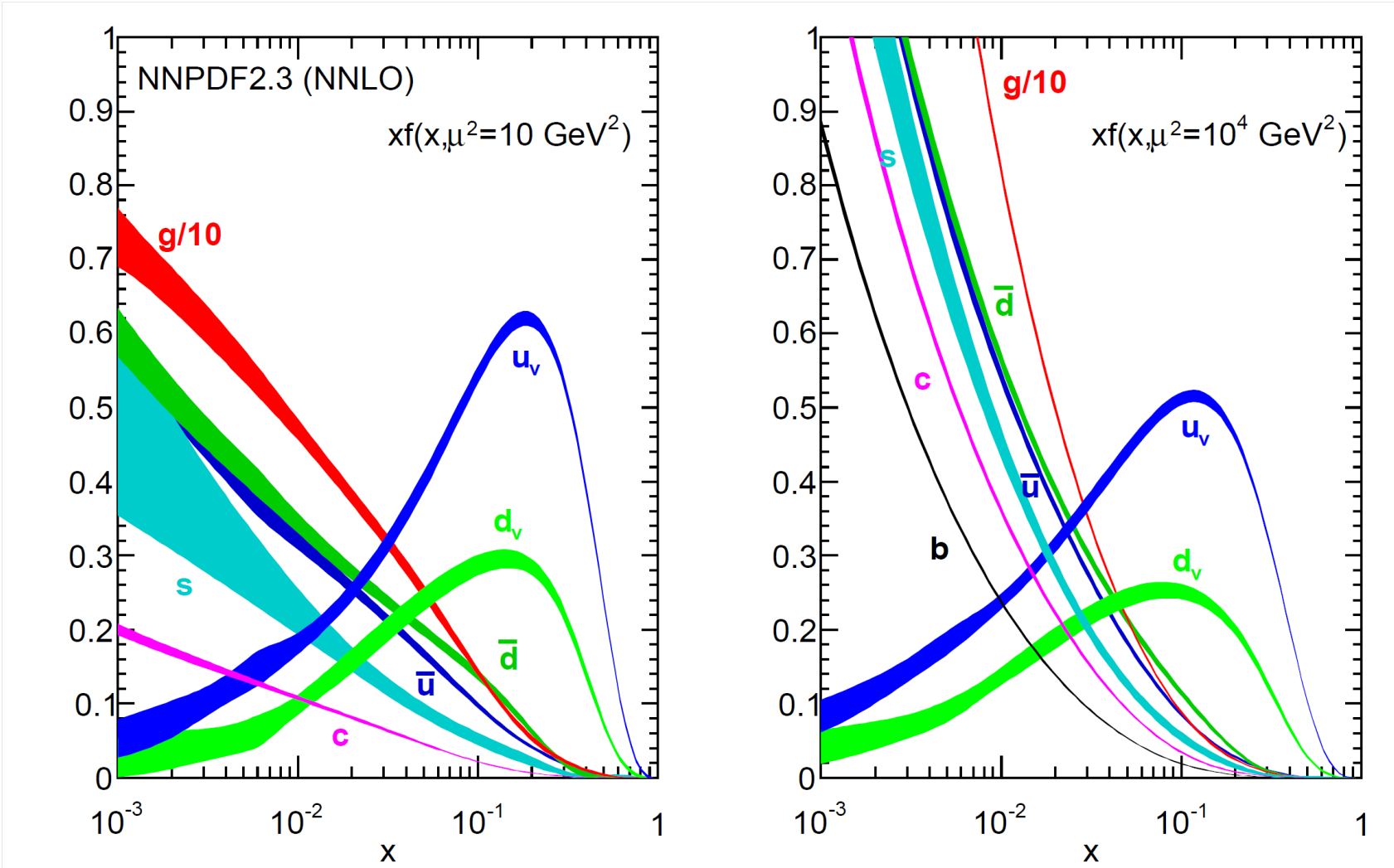
$$s_v(x) = \bar{u}_v = \bar{d}_v = \bar{s}_v = 0 \tag{96}$$

$$u_s(x) = \bar{u}_s = d_s = \bar{d}_s = s_s(x) = \bar{s}_s = S(x) \tag{97}$$

$$2F_1^{ep}(x) = \frac{1}{2}u_v(x) + \frac{4}{3}S(x) \tag{98}$$

$$2F_1^{en}(x) = \frac{1}{3}u_v(x) + \frac{4}{3}S(x) \tag{99}$$

$$2F_1^{ep}(x) - 2F_1^{en}(x) = \frac{1}{6}u_v(x) \tag{100}$$



Source: K. A. Olive et al. (Particle Data Group), Chin. Phys. C 38, 090001 (2014), and 2015 update. Reproduced from PDG.

The momentum sum rule

If the quarks and antiquarks carried the whole nucleon momentum, one would have

$$\int_0^1 x\left(u(x)+d(x)+s(x)+\bar{u}(x)+\bar{d}(x)+\bar{s}(x)\right)dx = 1 \tag{101}$$

Let us notice that

$$\begin{aligned} * &= \int_0^1 \left(F_2^{ep}(x)+F_2^{en}(x)\right)dx && (102)\\ &= \int_0^1 dx x\left(\frac{4}{9}(u+\bar{u})+\frac{1}{9}(d+\bar{d})+\frac{1}{9}(s+\bar{s})\right) && (103)\\ &+ \int_0^1 dx x\left(\frac{4}{9}(d+\bar{d})+\frac{1}{9}(u+\bar{u})+\frac{1}{9}(s+\bar{s})\right) && (104)\\ &\approx \frac{5}{9} = 0.5555... && (105) \end{aligned}$$

The sea is concentrated at $x \approx 0$ and carries little momentum — we have dropped the sea contribution!

- Experimentally it is measured to be: $* = 0.28$: which means that almost 50% is not seen in E-M interactions. The discrepancy indicates that quarks and antiquarks do not carry the entire nucleon momentum. A substantial fraction is carried by gluons.